
SUMMARY

Strong background in natural language processing, information retrieval, information extraction, and machine learning. Focused on practical solutions that enhance productivity, accessibility, and real-world user experience.

PROFESSIONAL & RESEARCH EXPERIENCE**Adobe Inc.**

San Jose, USA

AI/ML Machine Learning Engineer Intern

Jun. 2026 - Sep. 2026

- Worked on **contextual asset recommendations** by adjusting lexical and style signals based on user behavior and project state.

University of Washington

Seattle, USA

Graduate Research Scholar, eScience Institute

Dec. 2025 - Jun. 2026

- Built a **CI pipeline** to evaluate **research software engineering agents** for performance, efficiency, and reliability.
- Developed LLMoxie, an **open-source scientific RAG/LLM research tool** for cloud and HPC systems.
- Led GAIA Agentic AI, an **open-source agentic software engineering pipeline** for cross-domain geohazard and climate research.

Graduate Student

Sep. 2024 - Present

- Developed an open-response **medical question answering benchmark** with Microsoft Health AI researchers. **GEM @ ACL 2026**
- Built a **lay summarization system for medical texts** by fine-tuning Llama 3 8B & 70B LLMs. **1st Place, BioNLP @ ACL 2025**
- Created a **legal hallucination benchmark** via adversarial multi-agent LLMs.
- Developed a tool to generate **emergency alerts in 15 languages** by fine-tuning and constraining Llama 3 1B LLM.
- Built a Chrome extension to **improve web accessibility** for people with visual impairments using Llama 4 Maverick.

NLP Research Lab, NHN Diquet | #1 search engine and chatbot provider in Korea

Seoul, South Korea

Senior Researcher, Foundational Technologies Team

Mar. 2022 - Jun. 2023

- Led **legal conversation summarization** using DL/RL to assist legal consultations, **adopted by the Korea Legal Aid Corporation**.
- Led a **content recommendation** system based on user behavior, serving **5M users**.
- Developed a self-service **MLOps platform** for managing ML models, abstracting complex infrastructure to **enable non-expert teams to train and deploy solutions with private data** on-site.
- Developed data security and NLP tools for **personal data extraction and masking**, and sentence completion, **used by the Korean National Police Agency** to ensure compliance and accelerate criminal investigations.

Associate Researcher, Foundational Technologies Team

Feb. 2019 - Feb. 2022

- Built **high-throughput language analyzers** (POS, NE, sentiment, speech acts) and **autocorrect** models (reduced errors by 47%), powering high-traffic search engines and chatbots (**millions of queries per day**).
- Developed a **recommendation system for course and job opportunity matching**, used by **35k university students**.
- Built a customizable, **hybrid NER** model using deep learning and pattern matching. **1st Place, U+ BMT NER**

Intelligent Systems Lab, Dong-A University

Busan, South Korea

Researcher

Dec. 2015 - Feb. 2019

- Conducted research on **machine reading comprehension** for question answering using language analysis. **Master's Thesis**
- Built **end-to-end DL language analyzers** for POS, NER, DP, and SRL. **Bronze Prize, Korean NLP Competition**
- Developed a hybrid **speech act classifier** fusing hand-crafted rules with DL. **Poster @ HCLT 2018**

EDUCATION**University of Washington**

Seattle, USA

M.S. in Computational Linguistics (AI / NLP) (GPA: 3.98 / 4.0)

Sep. 2024 - Present

Dong-A University

Busan, South Korea

M.S. (GPA: 3.91 / 4.0) & B.E. (GPA: 3.97 / 4.0) in Computer Engineering

Mar. 2014 - Feb. 2019

SKILLS

NLP • IR • IE • Text Mining • ML • Deep Learning (DL) • Parameter Efficient Fine-tuning (PEFT) • Supervised Fine-tuning (SFT) • Python • Java • C • C++ • Bash • TensorFlow • PyTorch • scikit-learn • NumPy • NLTK • spaCy • Regex • Git • CI/CD • Docker • SQL • Databases (MariaDB, MySQL, Oracle DB)